```
def _generate_typosquatting_variants() -> list[str]:
    replacements = {
        "a": "4", "e": "3", "i": "1", "o": "0",
        "l": "1", "m": "rn", "w": "vv",
    }
    suffixes = ["-secure.com", "-verify.net", "-login.org", "-support.com"]
    variants: set[str] = set()

    for brand in COMMON_BRANDS:
        lowered = brand.lower()
        for ch, repl in replacements.items():
            if ch in lowered:
                variants.add(lowered.replace(ch, repl, 1) + ".com")

        if len(lowered) > 3:
            variants.add(lowered[:3] + lowered[3] + lowered[3:] + ".com")

        variants.add(lowered + "s" + ".com")
        for suffix in suffixes:
            variants.add(lowered + suffix)

    return sorted(variants)

# Example output:
# "paypa1.com", "arnazon.com", "goog1e.com",
# "faceb00k.com", "netflix-verify.net", ...
```


```
{
  "timestamp_utc": "2026-06-12T14:02:31Z",
  "data": {
    "rows_used": 318280,
    "train_rows": 254624,
    "test_rows": 63656,
    "label_distribution": {
      "legitimate_0": 140540,
      "phishing_1": 177740
    }
  },
  "selection": {
    "metric": "phishing_f1",
    "best_model": "calibrated_linear_svc"
  },
  "models": [
    {
      "name": "calibrated_linear_svc",
      "accuracy": 0.9918,
      "precision_phishing": 0.9924,
      "recall_phishing": 0.9931,
      "phishing_f1": 0.9927,
      "macro_f1": 0.9915,
      "train_seconds": 41.2
    },
    {
      "name": "logistic_regression",
      "accuracy": 0.9901,
      "phishing_f1": 0.9909
    }
  ],
  "ranking": ["calibrated_linear_svc", "logistic_regression",
              "linear_svc", "sgd_classifier",
              "passive_aggressive", "multinomial_nb"]
}
```